

Run #133 — confirmation desk day: the first vision board

Dates: started 2026-08-29, completed 2026-08-30 — the leaderboard dates a board by when its run was created, so it files this one under the 29th **Workflow:** confirmation-desk-day , 9 steps / 33 checks, **uncalibrated par Contestants:** the four vision-tagged models, **2 trials each**, effort and output budget unpinned (32768 tokens, both wire protocols) **Run:** #133 on the live desk database · 8 clean trials · 0 invalid · 0 errors

Board

#	Contestant	Objective	OVR	GRD	ADH	SYN	PRC	EFF	CON	calls
1	deepseek-v4-flash-vision	86.3	81	94	99	99	60	24	96	46,30
2	glm-5-3-flash	87.9	78	89	99	99	70	18	86	63,37
3	gemini-3-7-flash	81.8	75	84	99	99	54	16	89	47,59
4	gpt-5-6-luna	84.8	68	59	99	86	94	10	89	59,74

Zero truncation, zero malformed tool calls, zero step errors across all eight trials.

The highest objective score does not win the board. glm-5-3-flash posts 87.9 to deepseek's 86.3 and still places second: grounding carries 0.32 of OVR and deepseek reads better (94 vs 89), consistently (CON 96 vs 86), in fewer calls. gpt-5-6-luna shows the same effect at the bottom — 84.8 objective, third-highest in the field, but last on OVR because its GRD is 59. **Read that 59 with the analysis below before reading it as weak sight:** three of its four grounding misses are one extraction-pipeline failure that hits gemini-3-7-flash identically, and the check that separates them rewards asserting an unverifiable number over honestly reporting an unobtainable one.

What this board was built to answer

Can GLM 5.3 Flash — z.ai's first multimodal model — work a confirmations desk against Gemini, GPT and DeepSeek? Yes, on correctness: it records the highest objective score in the field and ties the leaders on every designed vision trap it was given. It places second because it is neither the most consistent nor the leanest.

The more useful answer is that **the question is nearly obsolete at this tier.**

The designed vision traps are solved

Three of the four traps this workflow was built around are **saturated at 8/8** — every contestant, every trial:

Trap	Document	Truth	Rate
Image-only scan, no text layer	conf-04 GOOGL	strike 205.0	8/8
Struck-through and amended strike	conf-09 NVDA	1045.0, not the struck value	8/8
Barrier direction carried only by a ticked box	conf-10 AMZN	DOWN_OUT , 171.2	8/8
A term the document never states	conf-07 META	initial_price null, status invalid	8/8

That last row is the one worth pausing on. Step 7 asks for a value that is not in the document, with the strike sitting nearby as an obvious substitute. **No model substituted, in any trial.** The pilot board saw the incumbent substitute in 2 of 6 runs; on this field the behaviour is gone.

What still discriminates — 6 of 28 checks

Per-check pass-rate tally across the whole field. Keyed by label, so a check appearing in several steps is counted once per occurrence.

Check	Rate	What it is
Step 6 — ORCL notional 636,000	2/8	The hardest check. A substitution failure, not a reading one — see below.
<code>get_confirmation_batch</code> called	24/48	Procedure, not sight
Step 8 — <code>skipped_count</code> = 1	4/8	Booking restraint and counting
Step 3 — AMD strike 185.0	6/8	conf-08 page 2 — an extraction-pipeline failure, see below
Step 3 — AMD initial_price 178.9	6/8	Same document, same page, same cause
Step 9 — artifact names ARD-EQO-2026-04901	7/8	Carrying a reference into the summary

22 of 28 checks are dead (0/N or N/N). This tally is only valid because every arm on this board is healthy — a contaminated arm fails checks the healthy field passes and manufactures apparent discrimination. On the pilot board, with one arm poisoned by an infra defect, the same instrument read 8 checks dead where the repaired board read 25.

Neither hard check is an OCR failure

Both were re-examined against the transcripts and the source documents after the board was first published. Neither is what it looked like.

The ORCL notional is a SUBSTITUTION failure. The document states a Notional Amount of 636,000.00 in a low-contrast column; the generator also gives it `num_options: "4,000"` and a strike of 163.50. The three wrong answers are not misread digits:

Answer	Who	What it actually is
4,000	luna x2, gemini t1, glm x2	the contract count , reported as the notional
654,000	deepseek t1	computed 4,000 x 163.50
636,000	gemini t2, deepseek t2	correct — the stated value

Every contestant read the neighbouring strike 163.50 correctly, **8/8**. So the field can see that region of the page. What a faint value produces is not a wrong digit but a **substituted legible field** — the same instinct step 7 traps, minus step 7's explicit permission to answer null.

The AMD page-2 checks measure the EXTRACTION PIPELINE, not the agent. Because this workflow routes the extraction sub-call to the contestant, a contestant can fail here without ever misreading anything itself. conf-08's outcome is bimodal and perfectly split by model, identical on every one of their attempts:

- gpt-5-6-luna — terms: {}, always
- gemini-3-7-flash — terms: {}, always
- glm-5-3-flash, deepseek-v4-flash-vision — 185.0 / 178.9, always

Stage 1 succeeds for all four (family, counterparty and reference ARD-EQO-2026-04781 at confidence 0.99). **Stage 2 — filling the term schema from the rendered page — returns an empty dict for two of them.**

CORRECTION (2026-08-31): a cross-contestant leak, not a fabrication

An earlier revision of this report said gemini-3-7-flash "asserted the correct values without accessible evidence". **That was wrong, and the error is instructive.** The arena transcript records only the PARENT agent's tool calls; work done inside a task() subagent never appears in it. "No tool result contains the value" was therefore never valid evidence, and the trace database — which does record subagent spans — tells a different story.

What actually happened, in both gemini trials:

```
tool task      -> chain general-purpose      (subagent, invisible to the transcript)
tool read_file  /large_tool_results/call_3c5fd773d1f2455993d7552e
tool record_answer {"strike": 185, "initial_price": 178.9}
```

That file is not gemini's. It belongs to glm-5-3-flash — written by glm's own parse_trade_confirmation at 16:03, and read by gemini at 17:27 and again in its second trial. It contains conf-08's strike: 185.0, initial_price: 178.9.

/large_tool_results/ is a **content-addressed store that is written per session but read globally.** ContentAddressedFilesystemBackend._latest_artifact() and ls() filter only on kind == "tool_result" and the rendered path — there is no workflow_id, session_id or thread predicate — while capture_tool_result() writes both ids. The workflow-scoped list_artifacts / read_artifact tools are the documented recovery route; the filesystem backend is an unscoped second door to the same store, and glob over it lists every other session's ids.

So a later contestant can read an earlier contestant's answers. That biases a board by POSITION IN THE FIELD — the same failure class as leftover fixture rows making each successive match's name resolution harder, and just as silent.

Measured on this board:

Contestant	own conf-08 extraction	foreign /large_tool_results/ reads
<code>gpt-5-6-luna</code>	0 of 5 attempts	none
<code>gemini-3-7-flash</code>	0 of 6 attempts	glm's, in both trials
<code>glm-5-3-flash</code>	4 of 4	one of luna's (carried no useful value)
<code>deepseek-v4-flash-vision</code>	2 of 2	none

`gemini-3-7-flash` 's **step-3 grounding passes are contaminated**. Its own extraction of conf-08 never once succeeded; the values it recorded came from glm's data. Its GRD of 84 is therefore inflated, and on its own evidence it would have scored on those checks roughly where luna did. `glm-5-3-flash` and `deepseek-v4-flash-vision` are unaffected — their extractions succeeded on their own, every time. Fifteen further reads of this kind appear across runs #129-#132, so this is not unique to this board.

What this does NOT change: luna's GRD 59 is real and is not a sight deficit. The value `178.9` appears in **zero spans** of either luna thread — its extraction genuinely never produced it — and luna neither fabricated a number nor took one from a neighbour. It reported null and said why. Of the four contestants it is the only one that both failed the extraction and declined to obtain the answer by any other route.

One extraction failure, three of luna's four grounding misses

The cascade is why luna's GRD is an outlier rather than a gradient:

- conf-08 stage 2 returns `terms: {}` -> step 3 `strike` and `initial_price` recorded null (2 checks)
- conf-08 therefore validates `invalid` -> at step 8 luna skips META **and** AMD, so `skipped_count=2` rather than 1 (1 check)
- the fourth miss is the ORCL notional, which most of the field also misses

Luna's identical GRD of 59 on the pilot board is consistent with the same deterministic cause.

The board's variance is procedural, not visual

`glm-5-3-flash` scored **97.0 and then 78.8** — an 18-point spread on identical inputs, and the reason it dropped from first to second when its second trial landed.

Every point of that spread is one check. Its two trials read the documents *identically*: same strikes, same barrier, same absent term, same faint-notional miss. What differed is that trial 1 called `get_confirmation_batch` at all six graded steps and trial 2 called it at none.

`deepseek-v4-flash-vision` shows the same thing inverted. Its lower-scoring trial (81.8, 30 calls) is the one where it read **every trap correctly including the faint notional** — the only trial on the board to do so — and it scores lower than its 90.9 trial because it skipped the batch re-reads.

So the single widest-swinging check on this board rewards re-fetching a payload the model already holds. That is the concern the pilot report raised about these checks, now with a number on it: `get_confirmation_batch` **accounts for more objective variance than every vision check combined**. It measures procedure, and a model answering correctly from context it already has is arguably being efficient rather than wrong. This is the same class as the "step-scoped check penalises reading the evidence one step early" defect already recorded in `CLAUDE.md`.

Par is still not calibrated, and now there is evidence for why

`confirmation-desk-day` declares no `par_tool_calls`, so EFF runs on the legacy hyperbolic curve against a `designed_par` of 9 — the theoretical minimum. That crushes every EFF on this board into 10–24 on a 0–99 scale, and EFF is 0.16 of OVR.

The established way to calibrate is the median of the **fully-correct** trials. **This board produced none: not one of the eight trials passed every check.** The pilot's single 33/33 run is one data point, not a distribution. So par stays unset rather than being invented from a sample that does not exist, and **EFF and OVR here are not comparable with calibrated boards.**

For the record, the counted call distribution over 8 clean trials: median **53**, range **30–74**.

Why this board can be trusted

The pilot board found three harness defects, each of which cost a real contestant real points, and each invisible to the golden replay. All three are fixed here, and this run is the evidence:

1. **StreamChunkTimeoutError after 120s.** Step 1 parses six documents in one tool body — 12+ vision calls synchronously — so the outer stream emits nothing for minutes and langchain's default fires on a connection that is idle rather than dead. Launched with `LANGCHAIN_OPENAI_STREAM_CHUNK_TIMEOUT_S=900`; zero contestants died to it.
2. **Unaddressable upload paths.** A model told the files are at `/artifacts/uploads/confirmations/` and then rejected for using that spelling scored

30.3 with grounding 0/10 on the pilot. Fixed; zero path errors here.

3. A binary `read_file` poisons the conversation permanently. `deepagents` returns a binary read as a media content block; three of four gateways reject that shape from a *tool* message, each in its own dialect, and the rejected message stays in the history so every later turn redraws the same 400. On the pilot this took `deepseek-v4-flash-vision` from a real 100.0 to a recorded 36.4 and last place. `BinaryReadGuardMiddleware` now intercepts it in all three hand-built agent stacks **and** in the `general-purpose` subagent that `deepagents` adds for itself — a fourth stack that inherited the parent's full toolset while running unguarded and unaudited.

Zero errors across eight trials is the result of those three fixes, not of an easy workflow.

Caveats a reader must carry

- **Do not read this ordering as a vision ranking.** 22 of 28 checks are saturated, separation comes from consistency, procedure and volume — and the grounding axis itself is partly measuring the extraction sub-call rather than the agent, because this workflow routes that sub-call to the contestant.
- `gpt-5-6-luna` 's GRD 59 is not a sight deficit. It is one deterministic conf-08 extraction failure cascading into three checks, plus the field-wide notional. `gemini-3-7-flash` fails the same extraction 0 of 6 times and scores 84 anyway, because it read the values out of `glm-5-3-flash` 's tool result through an unscoped shared store.
- `gemini-3-7-flash` 's GRD is CONTAMINATED and its rank should not be trusted. See the correction above. The leak is a harness defect, not misconduct by the model — the store was reachable and the model used it — but the two step-3 grounding checks it passed are not evidence of its own reading. The other three contestants are unaffected on this check. Scores here are left exactly as measured rather than silently adjusted, in keeping with this arena's rule that a flag is a caveat on a measurement and never a retroactive penalty.
- **EFF is uncalibrated** (see above). Do not compare it across boards.
- **One arm was re-run.** `glm-5-3-flash` lost a trial to a transport-level `RemoteProtocolError` — the peer closed the connection mid-body — and infra trials are skipped rather than retried, which would have left the headline contestant at half the field's depth with no CON. Its `scored` row was deleted and the arm re-run to two fresh trials via `--resume 133` , at the same effort and the same re-supplied output budget. **The repair changed the podium:** on the single surviving trial GLM stood at OVR 85 and first place; at full depth it is 78 and second. Its objective mean is 87.9 either way — the coincidence is real and is why OVR, not the objective percentage, ranks the board.
- **Two trials, not five.** CON is measured but from a small sample.

- **The extraction sub-call routes to the contestant.** Without that, every model would read every document with one shared extractor and every vision check would land N/N carrying no signal.

Next

- **Reconsider the `get_confirmation_batch` checks.** They are 1 of the 6 surviving discriminators and the largest single source of objective variance, and what they reward is re-fetching held context.
- **Scope the CAS read path.** `_latest_artifact()` and `ls()` must filter by `workflow_id` / `session_id` the way `capture_tool_result()` already does when it writes. Until they do, `/large_tool_results/` is a cross-session read channel and every arena board is exposed to it — 15 such reads appear across runs #129-#132 alone. Then re-run the contaminated arm.
- **Fix the step-3 grounding check, which currently inverts the desk's own standard.** As written it scores a model for producing a number it cannot evidence and scores zero for reporting, correctly, that the term could not be extracted. Either grade the extraction outcome explicitly, or word the step like step 7 so that a justified null is a legal answer.
- **The workflow needs harder traps.** Its designed vision difficulty is solved by the whole field, and the two checks that still separate contestants turned out to measure substitution and an extraction failure rather than sight. That is where the next iteration should aim.
- **Do not calibrate par until a trial passes every check.** The instrument is the median of fully-correct trials, and that set is currently empty.